

Birzeit University
Faculty of Engineering & Technology
Computer Science Department
Second Semester 2025/2026
Discrete Math (Comp 233) First Exam
Section 9

Student Name:

Student ID:

Question One [10 Marks]:

Select the correct answer for each of the following:

1. If $n = 5k + 2$ for some integer k , what is the value of $n^2 \bmod 5$?
 - a. 0
 - b. 1
 - c. 4
 - d. 2
2. What is the largest power of 6 that divides $30^5 \cdot 40^3$?
 - a. 6^3
 - b. 6^5
 - c. 6^7
 - d. 6^9
3. Suppose m and n are integers with $m \bmod 6 = 3$ and $n \bmod 6 = 4$. What is $(m + n) \bmod 6$?
 - a. 7
 - b. 1
 - c. 0
 - d. 5
4. Which of the following is **NOT** equivalent to the statement: “ $\forall$ integers n , if n is divisible by 4, then n is divisible by 2”?
 - a. Every integer divisible by 4 is also divisible by 2.
 - b. There is no integer divisible by 4 that is not divisible by 2.
 - c. Being divisible by 4 is a sufficient condition for an integer to be divisible by 2.
 - d. Some integer divisible by 4 is divisible by 2.
5. Consider the statement: “ $\exists$ a real number x such that $x^2 + 1 = 0$ ”. Which of the following best describes this statement?
 - a. The statement is true; $x = 0$ satisfies it.
 - b. The statement is false; for every real number x , $x^2 \geq 0$, so $x^2 + 1 \geq 1 > 0$.
 - c. The statement is true; $x = -1$ satisfies it.
 - d. The statement is undefined since $x^2 + 1$ has no real solution.

Question Two [10 Marks]:

- A) Prove that for every odd integer n , $n^2 + 3$ is divisible by 4. [4 Marks]
- B) Prove that for all integers a , b , and c , if $a \mid b$ and $b \mid c$, then $a \mid c$. [3 Marks]
- C) Use the quotient-remainder theorem with $d = 3$ to prove that for every integer n , the integer $n^2 + 2n$ has the form $3k$ or $3k + 2$ for some integer k . [3 Marks]

Question Three [10 Marks]:

- A) Consider the following statement: “For all integers m and n , if m and n are both odd, then $m + n$ is odd”.
- Disprove the statement by providing a counterexample.
 - State the correct conclusion about $m + n$ when m and n are both odd, and prove it. [4 Marks total]
- B) Prove that for all rational numbers r and s , the number $r^2 + s^2$ is a rational number. [6 Marks]

Question Four [10 Marks]:

- A) Let S be the set of students at a university and C be the set of courses offered. Let $E(s, c)$ be “student s is enrolled in course c ”.
- Express the following statement symbolically using $\forall$, $\exists$, and the predicate E : “There is a course that every student is enrolled in”. [3 Marks]
 - Write the **negation** of the following statement, in a form where no negation symbol appears before a quantifier: $\forall s \in S, \exists c \in C$ such that $E(s, c)$. [3 Marks]
- B) Use a **truth table** to determine whether the following argument form is **valid**. Clearly indicate: [4 Marks]
- Which columns represent the **premises**
 - Which column represents the **conclusion**
 - Whether the argument is **valid or invalid**
 - A short explanation based on the truth table

Argument:

$$\begin{array}{l} p \vee q \\ p \rightarrow r \\ q \rightarrow r \\ \therefore r \end{array}$$

Good Luck